Zexuan (Pierson) Bian

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EDUCATION

Master of Engineering in Electronic Science and Technology, Changzhou University; Sep 2025 – present

- In progress; expected graduation: Jun 2028.
- Wang Zheng School of Microelectronics; supervisor: Prof. Mo Chen.

Bachelor of Engineering in Automation, Changzhou University; Sep 2021 – Jun 2025

RESEARCH INTERESTS

Nonlinear and chaotic circuits; memristors; neuromorphic circuits and computing; memristive reservoir computing.

RESEARCH EXPERIENCE

Research at Changzhou University; theoretical work conducted jointly with Prof. Mo Chen.

Memristive reservoir computing and temporal prediction; Oct 2026 – present

Ongoing numerical research

- Compared operating points across a Hopf bifurcation using matched inputs.
- Developed MATLAB workflows for one-step prediction and nonlinear system emulation.
- Evaluated temporal memory using state-reset controls and digital delay baselines.

DC-induced firing mechanisms and stimulus modulation; Jun 2026 – present

Theoretical and numerical research

- Compared firing responses across five initial conditions using MATLAB.
- Analyzed firing dynamics using dynamic route maps and Poincaré returns.
- Proposed fixed-current basin comparisons to study initial-condition dependence.

Periodically driven memristive neuron circuits; Mar – Sep 2026

Theory, circuit simulation, and PCB experiments

- Analyzed firing dynamics using MATLAB, finite-time Lyapunov analysis, and LTspice.
- Independently conducted PCB experiments and recorded firing waveforms.
- Wrote the original manuscript draft as first author.

MANUSCRIPTS & WORK IN PROGRESS

Firing dynamics and underlying mechanisms in a second-order memristive neuron circuit with periodic nonlinearity

Zexuan Bian, Zezhong Li, Huagan Wu, Quan Xu, and Mo Chen. (2026).

Under review.

Bifurcation dynamics and temporal prediction in a second-order memristive reservoir circuit

Manuscript in preparation.

Research topic: DC-induced firing mechanisms and stimulus modulation

Manuscript in preparation.

ORAL PRESENTATIONS

Firing dynamics and underlying mechanisms in a second-order memristive neuron circuit with periodic nonlinearity; 7 Nov 2026

The 19th International Workshop on Complex-systems for Future Technologies and Applications (IWCFTA 2026), Zibo, China.

Complex dynamics of a multi-equilibrium memristive circuit; Apr 2026

9th Annual Conference on Chaos and Nonlinear Circuits, Circuits and Systems Branch of the Chinese Institute of Electronics, Dalian, China.

SKILLS AND INTERESTS

Languages: Chinese—native; English—near-fluent.
Academic skills: Proficient in using agentic AI tools with MATLAB and Python to design computational research workflows and develop scientific software.
Hardware: Proficient in using EDA tools and related software for analog and digital circuit experiments.
Personal projects: Regularly build small tools and reusable agent skills through AI-assisted coding ("vibe coding").
Interests: Travel, photography, and exploring the humanities and history.